

C.T.

Software Engineer – Python / Web

London, UK

Profile

Early-career software engineer focused on Python, web development, and public-interest data. Founded an open-source civic technology project making UK wealth inequality data accessible and interactive. DevOps and Security Lead of a production-grade FastAPI platform with a 4-person international team. At GE Digital, integrated automated security scanning into ~72 AWS CI/CD pipelines and learned a legacy Smallworld/Magik codebase on the job. Published lead-author research in Springer proceedings and present technical ideas to non-technical audiences through university outreach work.

Experience

Open-Source Civic Technology Project

2026 – Present

Founder & Lead Developer

London, UK

- Building open-source tools to make UK wealth inequality data accessible and impossible to ignore, with the goal of informing public understanding of tax policy and economic structures.
- Created a FastAPI backend with security headers, rate limiting, GZip compression, and CSV download; deployed via GitHub Actions with 8 CI/CD pipelines.
- Built 11 reproducible Python data pipelines ingesting ONS, HMRC, BoE, and WID government datasets, backed by a 40+ source open-data registry with full citations.
- Developing a policy microsimulation prototype to model the potential impact of UK tax reforms with provenance tracking.
- Built 72 Vue 3 components including 13 interactive chart visualisations with TailwindCSS, dark mode, and WCAG AA accessibility; 1,235+ automated tests.

Collaborative Data Platform (pre-startup)

2025 – Present

DevOps & Security Lead

Remote (international team)

- Architected a Python/FastAPI backend with 105 ORM models, 180 database migrations, and 120+ REST API endpoints secured by JWT/RBAC auth gates, per-IP rate limiting, and Stripe billing.
- Led a security hardening sprint: identified and remediated 6 CRITICAL + 13 HIGH vulnerabilities (SSRF, SQL injection, XFF spoofing, IDOR, brute-force) across 20 PRs with adversarial review.
- Built 39,500+ automated tests across a 4-tier framework (CI/validation/golden/nightly) with property-based testing (Hypothesis) and 125 golden-master scenarios.
- DevOps and Security Lead in a 4-person team, reviewing and merging 95% of all pull requests across 640+ PRs and driving 23 structured development phases.

Middlesex University

Jun 2025 – Present

Outreach Contractor / Workshop Lead

London, UK

- Deliver talks and workshops for schools and prospective students; regularly explain technical ideas to non-technical audiences and represent the university externally.
- Serve as the team's go-to data person (beyond the job description): data cleaning, visualisation, insight generation, and building automation scripts in Python.
- Contribute to a podcast initiative with secondary schools exploring societal pressures on young people; now developing a workshop programme bringing data literacy into schools.

GE Digital

Jun 2023 – Sep 2024

Software Engineer Intern

Cambridge, UK

- Integrated automated Nmap scanning into ~72 AWS CI/CD pipelines in Jenkins using Bash and Groovy, reducing manual release checks and improving pre-release security coverage.
- Learned and maintained a legacy Smallworld/Magik codebase on the job (a language not previously encountered): fixed bugs, improved database interactions, added MUnit tests, and documented the system for onboarding.
- Worked with DevOps engineers and security architects on release-process hardening across a large enterprise environment.
- Received the GE Spotlight Award for “Deliver with Focus” for proactive ownership of a security-driven delivery initiative.

- Led research on the N-Person Iterated Prisoner's Dilemma, building a Python simulation framework; published lead-author paper in Springer proceedings (SGAI-AI 2025) and presented as conference speaker.

Selected Projects

Taskdeck

C#, .NET 8, Vue 3, SQLite

Local-first developer tool with proposal-first automation: every AI-generated change produces a structured diff requiring explicit approval before application. 188-endpoint REST API with real-time SignalR, multi-provider LLM orchestration, and 8,600+ automated tests with architecture-boundary enforcement at CI.

N-Person Prisoner's Dilemma Research Platform

Python

Simulation and analysis framework for multi-agent cooperation research, supporting experiments, visualisation, and publication-quality analysis.

EduHub

Vue.js, Node.js / Express, MongoDB

Full-stack application for discovering and booking after-school lessons, including search, filtering, cart/favourites flows, validation, and RESTful APIs.

Education

Middlesex University

Graduated Jun 2025

BSc (Hons) Computer Science with Industrial Placement, First Class Honours

London, UK

Relevant study included full-stack development, AI and algorithms, data science and machine learning, software engineering, operating systems, and computer networks.

Publication & Awards

Lead-author paper: "Navigating the N-Person Prisoners' Dilemma: From the Tragic Valley to the Collaborative Hill." *Artificial Intelligence XLII / SGAI-AI 2025*, Springer. Presenting conference speaker.

GE Spotlight Award – Deliver with Focus

Dec 2023

Recognised for leading an essential DevSecOps initiative that integrated automated port scanning into AWS-based CI/CD pipelines.

Leadership & Service

Middlesex University

Nov 2024 – Jul 2025

Student Voice Leader & Ambassador

London, UK

- Gathered and presented student feedback to faculty and departmental boards; represented the university at information sessions, campus tours, and open-day activities.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash, Groovy, C#, Java

Backend / Web: FastAPI, Node.js / Express, REST APIs, Vue 3, React, HTML/CSS, TailwindCSS

Infrastructure: Docker, AWS, Jenkins, GitHub Actions, CI/CD, Git, Linux

Databases / Data: PostgreSQL, MySQL, SQLite, MongoDB, Pandas, scikit-learn

Testing: pytest, Vitest, Playwright E2E, Hypothesis (property-based), MUnit

Also: Smallworld/Magik, D3.js, secure automation, OpenTelemetry